Notes: Cookson, Fox, Gil-Bazo, Imbet, and Schiller (JFE, 2026)

Social media as a bank run catalyst

Contents

1 Big picture

- **Question.** Does **social media** amplify classical **bank run risk**, beyond the traditional balance-sheet vulnerabilities that already make a bank fragile?
- **Setting.** The paper studies the March 2023 run on **Silicon Valley Bank (SVB)** and the broader distress among U.S. regional banks, combining **comprehensive Twitter data**, bank balance-sheet data, stock-market data, and deposit-flow data.
- **Core challenge.** It is hard to tell whether Twitter is merely reflecting bad fundamentals or actually amplifying a run. Banks with more attention on social media may also be larger, more salient, or already viewed as risky. The paper therefore needs to separate **pre-existing exposure to social media** from **classical run risk**, and then study timing carefully.
- **Main contribution.** The paper combines three layers of evidence:
  1. cross-sectional tests that relate **Twitter pre-exposure** before the run to **run-period stock losses**;
  2. cross-sectional tests that relate Twitter pre-exposure to **Q1-2023 deposit outflows**;
  3. high-frequency hourly and tweet-level tests that distinguish the role of **Twitter attention** from **Twitter sentiment**.
- **Why it matters.** Modern bank runs can unfold through digital banking and rapid online communication. If social media is an important coordination device, then traditional models of bank fragility remain right in spirit, but the speed and propagation mechanism are different.
- **Main empirical question.** Are banks that were more exposed to Twitter before the SVB collapse more vulnerable during the run, especially when they also had large **uninsured deposits** and large **mark-to-market losses**?
- **Main empirical results.**
  - A one-standard-deviation increase in Twitter pre-exposure predicts about **4.3 percentage points** more stock-market loss during the SVB run.
  - Twitter pre-exposure **interacts positively** with classical run-risk variables—especially % uninsured deposits and mark-to-market losses—to predict more severe stock declines.
  - The same interaction also predicts larger **Q1-2023 deposit outflows**, especially for **uninsured deposits**.
  - At the hourly frequency, high recent Twitter attention predicts more negative returns **only for high-run-risk banks**; for low-run-risk banks, the same relation is absent.
  - High-frequency tweet-level tests show that **run-relevant tweets**—especially those by the **tech community** and tweets about **running** or **contagion**—move bank stock prices immediately.
  - By contrast, **negative Twitter sentiment does not amplify bank run risk**. The paper's core mechanism is about **attention and coordination**, not simply negativity.
- **Economic takeaway.** Social media appears to act as a **bank run catalyst**: it does not replace classical balance-sheet risk, but it can amplify it by increasing attention, communication, and likely coordination among depositors.

2 Data and measurement (key objects)

2.1 Twitter data and text measurement

- **Main source.** The authors use academic-access Twitter API data.
- **Main sample window.** The raw Twitter collection spans **January 1, 2020 to March 14, 2023**.
- **Tweet sample.** The main sample keeps **original English tweets** containing a bank cashtag (e.g. $SIVB) for depository institutions with SIC codes 602, 603, or 609.
- **Sample size.**
  - **5,399,740** original tweets about banking firms from 2020 through March 2023.
  - **765,224** retweets from January 1 to March 31, 2023 in a separate retweet sample.
  - **544,888** unique Twitter users who authored at least one sample tweet.
- **Tweet-level variables.** The data include tweet text, timestamp, author identifier, and retweet counts.
- **Author-level variables.** The paper collects user profile descriptions, account age, and follower counts.
- **Sentiment.** The paper scores each tweet using **VADER** (Valence Aware Dictionary and sEntiment Reasoner), yielding a score on $[-1, 1]$ and separate positive and negative components.
- **Context dictionaries.** The authors build dictionaries for:
  - **balance-sheet** tweets,
  - **run-behavior** tweets,
  - **contagion** tweets,
  - and **tech community** users (based on profile words such as venture capital, founder, entrepreneur, startup).
- **Why the tech-community dictionary matters.** A key claim of the paper is that likely SVB depositors were highly concentrated in the tech/startup ecosystem, which is also highly active on Twitter.

2.2 Bank sample, balance sheets, stock prices, and deposits

- **Bank universe for the main cross-section.** The Twitter sample initially covers **359** banks with nonzero pre-period Twitter attention. Merging with intraday stock data leaves **335** banks; merging with balance-sheet risk variables leaves a final cross-sectional sample of **277** publicly traded banks and bank holding companies.
- **Stock price data.** Intraday stock prices come from **FirstRate Data**. These are used to measure:
  - run-period stock losses,

  - hourly returns during the SVB run,
  - and minute-level returns around tweet timestamps.

- **Run-period loss measure.** The main return outcome is the percentage stock-value loss from **March 1 to March 14, 2023**.
- **Deposit-flow data.** Deposit outflows are measured using **FDIC Call Reports**, comparing Q4-2022 to Q1-2023 deposits, separately for **uninsured deposits** and **total deposits**.
- **Uninsured deposits.** The variable **% Uninsured** is computed from deposits above and below the FDIC insurance threshold of $250,000.
- **Mark-to-market balance-sheet risk.** The paper constructs a market-based measure of unrealized asset losses by mapping banks' maturity exposures into Treasury-index price changes over 2022, with an adjustment for mortgage-backed securities.
- **Core run-risk measure.** The paper defines

$$\text{RunRisk}_i = \%\text{Uninsured}_i \times \%\text{LossMTM}_i.$$

  This is the paper's summary measure of balance-sheet fragility.

2.3 Pre-run and run-period definitions

- **Pre-run period.** January 1, 2023 through February 15, 2023.
- **Run-period tweet window.** Most run-period tweet counts are built over roughly **March 8 through March 13**.
- **Hourly panel window.** March 6 through March 10, 2023, with the run onset defined at **March 9 at 9 a.m.**
- **Tweet-level event window.** For each tweet, the return is measured from the last trade just before the tweet to the first trade in the **5-to-15 minute** window after the tweet.

2.4 Key descriptive facts

- **Twitter attention rises sharply during the run.** On a 30-day equivalent basis, banks average about **537** tweets in the pre-run period and about **2,279** during the run period.
- **Twitter attention is persistent.** The correlation between pre-run tweet counts and in-run tweet counts is about **0.85**. This is central to the interpretation of pre-exposure as a proxy for future attention capacity.
- **Pre-exposure is not just hidden run risk.** Twitter pre-exposure is only weakly correlated with pre-period tweet sentiment (about $r = 0.18$) and essentially uncorrelated with run risk (about $r = 0.02$).
- **SVB dominates the run conversation.** During the run period, SVB receives roughly **6,528** *run* tweets and **9,662** *contagion* tweets, far above the rest of the sample.

- **Investor tweets lead general discussion.** Tweets using the ticker **SIVB** spike before generic mentions of **SVB** or **Silicon Valley Bank**, suggesting that investor/depositor discussion led broader public discussion.
- **Tech-community tweets matter.** Tweets by accounts associated with founders/startups/VCs spike distinctly during the run and are more likely to mention the bank directly rather than the stock ticker, consistent with actual depositor coordination.

2.5 Unobservables and timing

- **Cross-sectional concern.** Banks with more Twitter attention may also differ in size, media coverage, analyst following, valuation, deposit franchise, or prior returns.
- **High-frequency concern.** Tweets during the run may react to price moves rather than cause them.
- **Paper's solution.** The design uses:
  - **pre-run** Twitter exposure for the cross-section,
  - **lagged** Twitter attention in the hourly panel,
  - and **narrow windows around tweet timestamps** for the minute-level tests.

3 Models and empirical specifications (all equations + interpretation)

3.1 Mark-to-market loss construction and run risk

The paper first builds a bank-level proxy for unrealized losses from duration exposure:

$$\%\text{LossMTM}_i \approx \sum_m \text{AssetExposure}_{im} \times \Delta\text{TreasuryPrice}_m,$$

where m indexes maturity or repricing buckets, with an additional adjustment for mortgage-backed securities.

It then defines the main balance-sheet risk variable as

$$\text{RunRisk}_i = \%\text{Uninsured}_i \times \%\text{LossMTM}_i.$$

Interpretation. The paper's core idea is that a bank is especially fragile when it combines a large uninsured deposit base with large unrealized asset losses. The social-media question is whether Twitter amplifies the effect of this classical fragility measure.

3.2 Determinants of Twitter pre-exposure

To study what predicts social-media exposure before the run, the paper estimates:

$$\text{TwitterPreExposure}_i = a + b_1\,\text{Size}_i + b_2\,\%\text{LossMTM}_i + b_3\,\%\text{Uninsured}_i + b_4 X_i + b_5 W_i + b_6 Z_i + \varepsilon_i.$$

Here:

- $\text{TwitterPreExposure}_i$ is the log number of tweets about bank i in the pre-run period,
- X_i collects information-environment controls such as analyst coverage and news coverage,
- W_i collects market controls such as prior returns, FOMC-event returns, and Google search attention,
- Z_i collects deposit and balance-sheet controls.

Interpretation. This is not the main causal equation. Instead, it is a credibility exercise. The authors show that Twitter pre-exposure is strongly related to bank **size** and to some general salience variables, but only weakly related to the fundamental run-risk variables. That supports using pre-exposure as a measure of social-media exposure rather than latent fragility.

3.3 Cross-sectional regression for run-period stock losses

The main stock-loss specification is

$$\text{StockLoss}_i = \beta_1\big(\%\text{LossMTM}_i \times \%\text{Uninsured}_i \times \text{PreExp}_i\big) + \text{lower-order terms} + \gamma X_i + \varepsilon_i.$$

Where:

- StockLoss_i is the percentage stock-value loss during the SVB run,
- PreExp_i is Twitter pre-exposure,
- X_i includes size, valuation, attention, prior stock returns, and deposit-franchise controls.

Interpretation. The triple interaction is the key coefficient. A positive coefficient means that social-media exposure **amplifies classical bank-run risk**: banks with both high uninsured deposits and large mark-to-market losses suffer bigger stock losses when they also had more Twitter pre-exposure.

3.4 Cross-sectional regression for deposit outflows

The deposit-flow specification mirrors the stock-loss specification:

$$\text{DepositOutflow}_i = \beta_1\big(\%\text{LossMTM}_i \times \%\text{Uninsured}_i \times \text{PreExp}_i\big) + \text{lower-order terms} + \gamma X_i + \varepsilon_i.$$

The dependent variable is the percentage loss of deposits from Q4-2022 to Q1-2023, estimated separately for **uninsured deposits** and **total deposits**.

Interpretation. If the same interaction that predicts stock-market losses also predicts actual deposit outflows, then stock-price evidence is more plausibly capturing run severity rather than unrelated equity-market news.

3.5 Hourly panel regression for lagged Twitter attention

For bank i in hour t, the paper estimates

$$\begin{aligned} r_{it} = a &+ b_1\, 1(t \geq \text{Mar 9, 9am}) + b_2\, 1(\text{TweetsHigh}_{i,t-1}) \\ &+ b_3\, 1(t \geq \text{Mar 9, 9am}) \times 1(\text{TweetsHigh}_{i,t-1}) + \eta X_{i,t-1} + \delta_i + \gamma_t + \varepsilon_{it}. \end{aligned}$$

Here:

- r_{it} is the hourly return in basis points,
- $1(\text{TweetsHigh}_{i,t-1})$ indicates whether tweet volume in the prior 4 hours is above the median,
- δ_i are bank fixed effects,
- γ_t are day-by-hour fixed effects,
- $X_{i,t-1}$ includes lagged stock returns and lagged news coverage.

Interpretation. This regression uses **lagged attention** to address timing. The key interaction asks whether recent Twitter attention predicts worse returns *after* the run begins, and whether this only happens for high-run-risk banks.

3.6 Tweet-level high-frequency return regression

Around each tweet timestamp, the paper estimates

$$\Delta p_{it} = a + b\,\text{VADERPos}_{it} + c\,\text{VADERNeg}_{it} + \eta X_{it} + \gamma_i + \varepsilon_{it},$$

with additional interactions between sentiment and indicators for:

- tweets authored by **tech community** users,
- **run** tweets,
- **contagion** tweets,
- and banks with **high run risk**.

The dependent variable Δp_{it} is the change in log price from the last trade just before the tweet to the first trade in the 5-to-15 minute window after the tweet.

Interpretation. These tests isolate the immediate price effect of tweet content. They are especially useful for asking whether negative sentiment itself drives price reactions, or whether the economically important force is tweet **attention and relevance**.

3.7 Attention versus sentiment

The empirical design makes a sharp distinction:

- **Attention** is measured by tweet counts or high/low tweet-activity indicators.
- **Sentiment** is measured by VADER positive and negative components.

Interpretation. This distinction is one of the main conceptual contributions of the paper. The authors show that **attention** amplifies bank run risks, whereas **sentiment** does not.

4 Identification and instruments (what makes the estimates credible)

4.1 What would be unconvincing in a simpler approach

- A simple correlation between tweets and losses would be weak evidence because tweets might just reflect public news or falling prices.

- Likewise, heavily discussed banks may simply be larger, more followed, or more vulnerable ex ante.
- During the run itself, contemporaneous tweet counts could be endogenous to returns.

4.2 What identifies the cross-sectional evidence

- **Pre-period design.** Twitter exposure is measured *before* the run, over January 1 to February 15, 2023.
- **Content validation.** The pre-run period contains little direct discussion of bank distress; the paper emphasizes that pre-period attention is not driven by bank-run talk.
- **Persistence in attention.** Pre-run exposure strongly predicts in-run attention (correlation about 0.85), making it a meaningful proxy for a bank's capacity to become focal on Twitter once a run starts.
- **Weak relation to run fundamentals.** Pre-exposure is almost uncorrelated with the paper's run-risk measure and only weakly related to pre-run tweet sentiment.
- **Rich control set.** The main regressions control for bank size, analyst coverage, news coverage, prior stock returns, deposit beta, branch footprint, market-to-book ratio, deposit flows, CRE exposure, liquid assets, and deposit concentration.

4.3 What identifies the high-frequency evidence

- **Lagged Twitter attention.** In the hourly regressions, tweet activity is measured over the previous 1, 2, or 4 hours.
- **Narrow event windows.** In the tweet-level tests, returns are measured only in the few minutes after a tweet.
- **Controls for recent prices and news.** The tweet-level regressions condition on stock-price changes and RavenPack news measures from the 10 minutes before the tweet.
- **Fixed effects.** Bank fixed effects absorb time-invariant differences across banks; day-by-hour fixed effects absorb common run dynamics.

4.4 Why the results point to coordination rather than generic negativity

- If Twitter were just a reflection of bad news, one would expect **negative sentiment** to be the key force.
- Instead, the hourly results show that **attention** matters for high-run-risk banks, while sentiment does not amplify run risk.
- The immediate price effects are strongest for **run-relevant tweets** and **tech-community tweets**, which is exactly what one would expect if depositor communication is the mechanism.
- The effect is stronger when tweets are **retweeted heavily**, reinforcing the idea that the social propagation of attention matters.

4.5 Robustness and external credibility

- The results survive using alternative pre-periods going back to October 2022.
- They remain similar when excluding **too-big-to-fail** banks.
- They remain similar when excluding **SVB** from the hourly sample, though magnitudes weaken somewhat.
- The balance-sheet loss measure is checked against fair-value disclosures in 10-K filings.
- A specification-curve exercise shows that the triple interaction between pre-exposure and run risk remains positive across a large number of specifications.

4.6 Relation to the literature

- **Bank run theory.** The paper is closest in spirit to Diamond–Dybvig style fragility and Goldstein–Pauzner style coordination models.
- **2023 banking-stress literature.** It complements recent work emphasizing uninsured deposits, mark-to-market losses, commercial real estate exposure, and flights to safety.
- **Social media and finance.** It extends the financial-social-media literature from asset-pricing effects to a setting where online communication can help drive a real financial crisis mechanism.

5 Tables and figures

5.1 Table 1, Figure 1, and Figure 2: dictionaries and the anatomy of the SVB conversation

Table 1
Contextual dictionaries for classifying types of tweets.

Balance sheet	Run behavior	Contagion	Tech community
duration	*run*	*systemic*	VC
cover & cash	*withdraw*	*spillover*	entrepreneur
mortgage backed securities	deposit money	fed	start up
mismatch	access accounts	regulator	startup
long maturity	pull & out	#contagion	founder
maturity mismatch	*get & out*	backstop	_venture
marked to market		whole system	venture capital
mark to market		spreading	
portfolio management		sparks	
liquidity		broader effects	
insured deposits		financial system	
MBS		meltdown	
hold to maturity		*contagion*	
HTM			
portfolio of loans			
liquidity management			
uninsured deposits			
balance sheet			

This table presents the keyword dictionaries for the major content dictionaries that we employ to classify tweets in to "Run," "Contagion,", "Balance Sheet," as well as "Tech Community" authors. For each of our content dictionaries we provide the seed words used, in the iterative creation process, in italics.

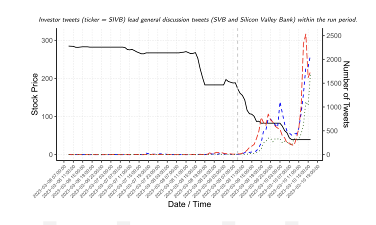


Fig. 1. *Silicon Valley Bank Stock Price and Twitter Mentions.* This figure shows the stock price of Silicon Valley Bank (SIVB) (black solid line), as well as the number of tweets mentioning "SIVB" (blue dashed line), "SVB" (red long-dashed line), and "Silicon Valley Bank" (green dotted line), over the period from March 06, 2023 at 9am (market open) to March 10, 2023 at 4pm (market close). The stock price is indicated on the left y-axis and the number of Tweets is indicated on the right y-axis. The gray vertical dashed line indicates March 09, 2023 at 9am.

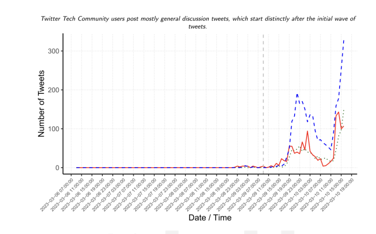


Fig. 2. *Tech Community Tweets about Silicon Valley Bank.* This figure shows the number of tweets mentioning "SIVB" (blue solid line), "SVB" (red dashed line), and "Silicon Valley Bank" (green dotted line), over the period from March 06, 2023 at 9am (market open) to March 10, 2023 at 4pm (market close). The sample includes only tweets posted by users who are part of the tech community as defined by corresponding dictionary of terms in the Twitter user biography. The gray vertical dashed line indicates March 09, 2023 at 9am.

- **Read.** Table 1 defines the main dictionaries for balance-sheet, run, contagion, and tech-community content.
- **Figure 1.** Investor-centric ticker tweets about **SIVB** spike before broader mentions of **SVB** or **Silicon Valley Bank**. This is important because it suggests the initial conversation came from financially engaged users rather than from generic news chatter.
- **Figure 2.** Tech-community tweets spike prominently during the run, and they focus directly on the bank. This supports the interpretation that likely depositors were active in the online conversation.

- **Punchline.** The socially relevant Twitter conversation is not random noise. It is tightly tied to the bank-run narrative and to the likely depositor base.

5.2 Figures 3–6 and Table 3: pre-exposure construction and validation

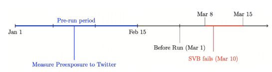


Fig. 3. *Timeline of SVB Run and Illustration of Empirical Strategy.* This figure presents a timeline that spans our main sample period from January 1 through March 15. It also illustrates definitions we use throughout the paper for pre-run period (January 1 – February 15) and run period (March 8 – March 15). Our main cross-sectional empirical strategy measures pre-exposure to Twitter by the number of original tweets authored about the bank's in the pre-run period and relates this pre-exposure to bank stock declines from March 1 (before the run) to the end of the run period. Figures A.5 and A.6 validate that tweet volume in the pre-period contains scant forward looking information in either its content or retweet patterns. Table A.4 Panels b and c, show robustness to alternative pre-periods.

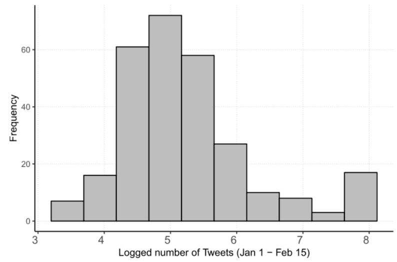


Fig. 4. *Distribution of Twitter Pre-Exposure.* This figure presents a histogram, depicting the cross-sectional distribution of Twitter pre-exposure. Twitter pre-exposure is the natural log of the number of cashtagged tweets about bank *i* posted between January 1, 2023 and February 15, 2023. The distribution is winsorized at the 95th percentile.

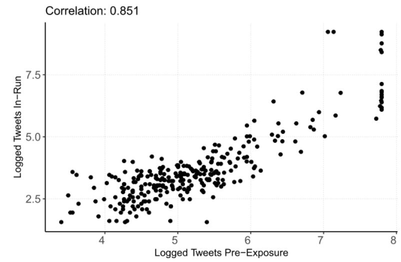


Fig. 5. *Twitter Exposure Pre- vs In-Run Period.* This figure presents a scatterplot that relates the logged number of Tweets in the pre-run period (January 1 – February 15) to the logged number of tweets in the run period (March 6 – March 14). As in our main Twitter pre-exposure measure, both distributions are winsorized at the 95th percentile. The correlation between these two variables is 0.85.

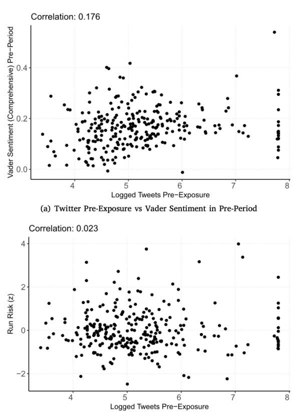


Fig. 6. *Twitter Pre-Exposure vs Tweet Sentiment and Run Risk.* This figure presents scatterplots depicting the relation between Twitter pre-exposure (x-axis) and either VADER sentiment of tweets in the pre-period (Panel (a)) or 'run risk' (Panel (b)) where run risk is % *Uninsured* × % *Loss MTM*, measured in Q4 2022. The correlation in Panel (a) is 0.18 and it is 0.02 in Panel (b).

Table 3
Determinants of twitter pre-exposure.

	Social media pre-exposure					
	(1)	(2)	(3)	(4)	(5)	(6)
ln(MCap) (z)	0.767***	0.730***	0.738***	0.744***	0.691***	0.592***
	(0.043)	(0.047)	(0.054)	(0.050)	(0.074)	(0.077)
% Loss MTM (z)	−0.027	−0.025	−0.026	−0.066*	−0.085**	−0.070*
	(0.040)	(0.039)	(0.039)	(0.037)	(0.037)	(0.036)
% Uninsured (z)	−0.051	−0.056	−0.055	−0.074	−0.067	−0.030
	(0.050)	(0.051)	(0.051)	(0.046)	(0.046)	(0.048)
Log N Articles (z)		0.101**	0.111*	0.087	0.045	0.037
		(0.043)	(0.058)	(0.053)	(0.054)	(0.052)
Log N Analysts (z)			−0.019	−0.054	0.004	0.015
			(0.071)	(0.072)	(0.075)	(0.070)
Ret 2022-Q4 (z)				−0.010	−0.009	0.009
				(0.042)	(0.041)	(0.040)
Ret 2022-Q3 (z)				−0.078*	−0.075**	−0.064*
				(0.040)	(0.038)	(0.038)
FOMC 2022 CAR (z)				−0.183***	−0.170***	−0.125***
				(0.050)	(0.045)	(0.047)
Δ SVI Q4-Q3 (z)				0.052	0.043	0.047
				(0.047)	(0.049)	(0.046)
% Dep Flow 2022-Q4 (z)					−0.042	0.103
					(0.053)	(0.085)
N ZIPs Branches (z)					0.064	0.063
					(0.044)	(0.043)
Deposit Beta (z)					−0.025	−0.035
					(0.038)	(0.037)
MTB BHC (z)					−0.083**	−0.036
					(0.034)	(0.034)
% CRE Loans (z)						−0.141***
						(0.041)
% Liquid Assets (z)						0.061
						(0.058)
Deposit Concentration (z)						0.071***
						(0.019)
Constant	5.235***	5.235***	5.235***	5.234***	5.234***	5.223***
	(0.038)	(0.038)	(0.038)	(0.036)	(0.035)	(0.035)
Observations	277	277	277	277	277	274
R^2	0.588	0.597	0.597	0.643	0.654	0.677
Adjusted R^2	0.583	0.591	0.590	0.631	0.637	0.657

This table presents OLS regressions that relate Twitter Pre-Exposure to characteristics of the bank, market and information environment that could drive attention to banks. The dependent variable is 'Twitter pre-exposure', i.e., the log of the number of tweets about a given bank in the pre-run period from Jan 1 through Feb 15, 2023. The explanatory variables include bank characteristics such as size (ln(MCap)) and valuation (MTB), run risk variables (% Loss MTM and % Uninsured depositors), attention characteristics (analyst coverage and number of newspaper articles in 2022, change in Google Search Volume SVI between Q4 and Q3 of 2022), stock returns (in Q3 and Q4 of 2022 and around 2022 FOMC meetings), measures of the deposit franchise (i.e., deposit beta from Drechsler et al. (2017) and the number of unique states with bank branches), and measures of deposit characteristics (% of commercial-real-estate (CRE) loans, deposit flow in Q4-2022, supply-driven deposit inflows in 2022, and deposit concentration). All variables denoted with '(z)' are standardized to have a mean of zero and standard deviation of one to ease magnitude comparisons.

* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.

- **Read.** Figure 3 lays out the timing of the empirical strategy: pre-exposure is measured before the run, then related to run-period outcomes.
- **Figure 4.** The distribution of pre-exposure is well behaved once winsorized.
- **Figure 5.** Pre-run and in-run tweet activity are very tightly related (correlation about **0.85**).
- **Figure 6.** Pre-exposure is only weakly related to pre-period tweet sentiment ($r \approx 0.18$) and essentially unrelated to run risk ($r \approx 0.02$).
- **Table 3.** The strongest predictor of pre-exposure is **size**; traditional run-risk variables are weakly related to it. This is crucial for the paper's identification argument.

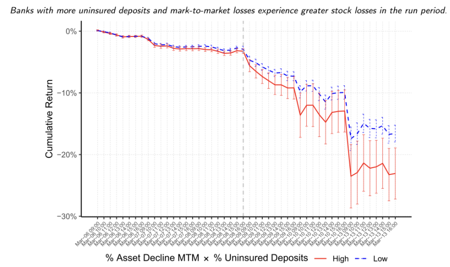


Fig. 7. *Cumulative Returns of Banks with High versus Low Run Risk.* This figure shows the average cumulative returns of US publicly listed banks and bank holding companies (BHCs) from March 06 through March 13, 2023. The sample is split by "Run Risk" defined as the % of uninsured deposits × the % MTM-implied asset value decline. 'High' represents the top half above the median (red solid line), 'Low' represents the bottom half (blue dashed line). The gray vertical dashed line indicates March 09, 2023 at 9am. 90% confidence intervals around the mean cumulative return are constructed by bootstrapping with clustering at the bank level.

5.3 Figure 7 and Table 4: stock-market evidence on amplification

Table 4
Twitter pre-exposure and bank stock losses.

Panel A: Baseline result

	% Stock value loss during run			
	(1)	(2)	(3)	(4)
% Uninsured (z)	4.249***		3.276***	2.715***
	(1.026)		(0.840)	(0.524)
% Loss MTM (z)	1.192		1.579*	0.227
	(0.879)		(0.814)	(0.499)
% Uninsured (z) × % Loss MTM (z)	1.742**		1.838**	0.778**
	(0.824)		(0.737)	(0.375)
Pre-Exposure (z)		4.317***	3.683***	3.724***
		(0.936)	(0.807)	(0.730)
Pre-Exposure (z) × % Uninsured (z)				2.695***
				(0.701)
Pre-Exposure (z) × % Loss MTM (z)				0.329
				(0.826)
Pre-Exposure (z) × % Uninsured (z) × % Loss MTM (z)				2.407***
				(0.561)
Constant	16.555***	16.112***	16.585***	15.666***
	(0.648)	(0.547)	(0.595)	(0.457)
Observations	277	277	277	277
R^2	0.190	0.185	0.312	0.430
Adjusted R^2	0.181	0.182	0.302	0.415

(*continued on next page*)

Table 4 (*continued*).

Panel B: Including controls for size, valuation, attention, prior stock returns, deposit flows, deposit franchise, risk

	% Stock value loss during run			
	(1)	(2)	(3)	(4)
Pre-Exposure (z) × % Uninsured (z) × % Loss MTM (z)			1.437***	1.660***
			(0.354)	(0.424)
Pre-Exposure (z) × % Uninsured (z)			1.862***	1.641***
			(0.530)	(0.613)
Pre-Exposure (z) × % Loss MTM (z)			−1.080*	−0.276
			(0.563)	(0.632)
Pre-Exposure (z)		1.922**	2.339***	1.649**
		(0.744)	(0.608)	(0.643)
% Uninsured (z) × % Loss MTM (z)	1.794**		0.541	0.710*
	(0.726)		(0.390)	(0.396)
% Uninsured (z) × % Loss MTM (z) × ln(MCap) (z)			1.627***	1.457***
			(0.400)	(0.404)
% Uninsured (z)	2.224***		2.306***	2.471***
	(0.664)		(0.471)	(0.477)
% Uninsured (z) × ln(MCap) (z)			1.354**	1.381**
			(0.586)	(0.584)
% Loss MTM (z)	0.693		−0.677	−0.383
	(0.690)		(0.464)	(0.538)
% Loss MTM (z) × ln(MCap) (z)			2.408***	1.770***
			(0.643)	(0.617)
ln(MCap) (z)	1.657	1.308	−0.565	−0.458
	(1.048)	(1.379)	(0.867)	(0.860)
Log N Analysts (z)	1.644**	1.921**	1.998***	2.262***
	(0.739)	(0.761)	(0.726)	(0.724)
Log N Articles (z)	−0.260	−0.512	−0.460	−0.649
	(0.595)	(0.616)	(0.550)	(0.545)
Ret 2022-Q4 (z)	−1.659***	−1.928***	−0.754*	−0.411
	(0.527)	(0.649)	(0.423)	(0.428)
Ret 2022-Q3 (z)	−0.709	−0.581	−0.167	−0.259
	(0.588)	(0.662)	(0.446)	(0.443)
FOMC 2022 CAR (z)	−2.185***	−1.780***	−2.152***	−1.674***
	(0.552)	(0.551)	(0.499)	(0.530)
Δ SVI Q4-Q3 (z)	−0.696	−0.671	−0.980**	−0.847**
	(0.515)	(0.531)	(0.406)	(0.409)
% Dep Flow 2022-Q4 (z)	−2.429***	−2.253***	−2.193***	−0.200
	(0.797)	(0.688)	(0.558)	(0.636)
N ZIPs Branches (z)	−1.048*	−1.583**	−1.227***	−1.295***
	(0.623)	(0.764)	(0.418)	(0.413)
Deposit Beta (z)	−0.086	0.215	0.584	0.389
	(0.557)	(0.560)	(0.403)	(0.402)
MTB BHC (z)	−1.323***	−1.276**	−1.211***	−0.971**
	(0.502)	(0.529)	(0.411)	(0.426)
% CRE Loans (z)				−0.557
				(0.449)
% Liquid Assets (z)				0.780
				(0.705)
Deposit Concentration (z)				0.813*
				(0.485)
Constant	16.560***	16.115***	15.422***	15.385***
	(0.550)	(0.464)	(0.416)	(0.412)
Observations	277	277	277	274
R^2	0.487	0.430	0.671	0.635
Adjusted R^2	0.460	0.404	0.644	0.599

(*continued on next page*)

Table 4 (*continued*).

Panel C: Controlling for tweet activity during run period

	% Stock value loss during run					
	(1)	(2)	(3)	(4)	(5)	(6)
Pre-Exposure (z)	4.317***	0.431	−0.408	−1.376	−0.892	−0.129
	(0.936)	(0.516)	(0.589)	(1.110)	(0.617)	(0.618)
Contagion Tweets In-Run (z)		6.696***			3.104**	3.942***
		(1.333)			(1.362)	(1.140)
'Run' Tweets In-Run (z)			7.252***		3.594***	4.398***
			(1.374)		(1.302)	(1.203)
Tech-Comm. Tweets In-Run (z)				7.150***	1.339*	0.280
				(1.906)	(0.799)	(0.830)
Constant	16.112***	16.096***	16.121***	16.127***	16.112***	16.108***
	(0.547)	(0.434)	(0.435)	(0.483)	(0.426)	(0.343)
Observations	277	277	277	277	277	274
R^2	0.185	0.484	0.489	0.372	0.509	0.666
Adjusted R^2	0.182	0.481	0.485	0.368	0.502	0.643
Funding controls						✓
Market controls						✓
Deposits controls						✓
Attention controls						✓
Valuation controls						✓

This table presents OLS estimates of regressions that relate bank stock losses during the run period to Twitter pre-exposure, mark-to-market losses, and percentage of uninsured deposits (Panel A presents the baseline results). All variables denoted with '(z)' are standardized to have mean zero and standard deviation of one. Panel B is similar to Panel A but additionally controls for bank size and valuation, attention characteristics, stock returns, measures of the deposit franchise and measures of deposit characteristics. Variable definitions are as in Table 3, with full detail on their construction in Appendix Table A.1. In Panel C, total 'Run', 'Contagion' tweets during the run count the number of tweets with at least one word in the 'Run' and 'Contagion' dictionaries between March 8 and March 13. Tech community tweets are counted over the same period, but based on the number of tweets about the bank stock posted by users with at least one tech community term in their Twitter user description. Robust standard errors are reported in parentheses.

* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.

- **Figure 7.** Banks with high run risk suffer much larger cumulative stock losses, and the divergence begins at the onset of the SVB run on **March 9**.
- **Table 4, Panel A.** Without controls, a one-standard-deviation increase in Twitter pre-exposure predicts about **4.317 percentage points** more stock-market loss.
- **Table 4, Panel A.** The triple interaction between pre-exposure, uninsured deposits, and mark-to-market losses is about **2.407 percentage points** without controls.
- **Table 4, Panel B.** With controls, the triple interaction remains large and significant, about **1.44 to 1.66 percentage points** depending on specification.
- **Table 4, Panel C.** Once the regressions control for in-run **run**, **contagion**, and **tech-community** tweet counts, the coefficient on pre-exposure shrinks sharply. This supports the mechanism that pre-exposure matters because it generates more run-relevant attention during the crisis.

5.4 Table 5: deposit-flow evidence

- **Read.** Table 5 checks whether the same logic appears in actual deposit data.
- **Uninsured outflows.** The interaction between run risk and pre-exposure is positive and significant, with the triple interaction around **1.47** without controls and **1.78** with controls.

Table 5
Twitter pre-exposure and deposit flows.

	Deposit outflows Q1-2023 (%)			
	Uninsured		Total	
	(1)	(2)	(3)	(4)
% Uninsured (z) × % Loss MTM (z) × Pre-Exposure (z)	1.47**	1.78**	0.673	1.51***
	(0.746)	(0.890)	(0.536)	(0.476)
% Uninsured (z) × Pre-Exposure (z)	1.59	0.989	1.74**	0.897
	(0.993)	(1.03)	(0.697)	(0.547)
% Loss MTM (z) × Pre-Exposure (z)	1.79*	0.208	0.788	0.501
	(0.923)	(1.19)	(0.704)	(0.679)
Pre-Exposure (z)	0.973	2.03	0.974	0.512
	(1.12)	(1.29)	(0.968)	(0.839)
% Uninsured (z) × % Loss MTM (z)	−0.878	−1.24	0.112	−0.260
	(0.652)	(0.835)	(0.369)	(0.459)
% Uninsured (z)	3.31***	4.25***	1.37***	1.83***
	(1.17)	(1.42)	(0.502)	(0.542)
% Loss MTM (z)	0.609	0.081	0.829*	0.088
	(0.759)	(0.886)	(0.470)	(0.545)
ln(MCap) (z) × % Uninsured (z) × % Loss MTM (z)		0.683		0.201
		(0.896)		(0.580)
ln(MCap) (z) × % Uninsured (z)		2.23*		1.89**
		(1.35)		(0.728)
ln(MCap) (z) × % Loss MTM (z)		2.68*		0.804
		(1.55)		(0.778)
ln(MCap) (z)		−0.831		1.58
		(1.57)		(1.02)
Deposit Beta (z)		−0.010		−0.169
		(0.816)		(0.521)
N States Branches (z)		−2.72**		−1.83**
		(1.30)		(0.719)
FOMC 2022 CAR (z)		−1.08		−1.39**
		(0.806)		(0.565)
Ret 2022-Q4 (z)		1.30		0.655
		(0.969)		(0.584)
Ret 2022-Q3 (z)		−2.37**		−0.396
		(1.13)		(0.659)
Supply-driven Deposit Inflow 2020 (z)		1.46*		1.11**
		(0.805)		(0.452)
Dep Inflow Uninsured 2022-Q4 (z)		−1.30		−1.70*
		(1.76)		(1.01)
Dep Inflow Insured 2022-Q4 (z)		0.165		−1.30
		(1.66)		(1.07)
Constant	4.79***	4.46***	−0.915*	−1.29**
	(0.863)	(1.06)	(0.534)	(0.569)
Observations	275	262	275	262
R^2	0.088	0.247	0.086	0.399
Adjusted R^2	0.064	0.171	0.062	0.339
Attention controls		✓		✓
Other valuation controls		✓		✓

This table presents cross-sectional OLS regression estimates relating quarterly bank deposit flows to Twitter pre-exposure, *% Uninsured* and *% Loss MTM*. The dependent variable, 'Deposit outflows (%)' is the percentage change in bank deposits from the end of Q4-2022 to the end of Q1-2023, multiplied by −1 so that a higher number indicates larger outflows. Columns 1 and 2 use the uninsured deposits (above the FDIC threshold of $250K), columns 3 and 4 use all deposits to construct outflows. Columns 2 and 4 additionally include controls for bank valuation and outside attention, as well as controls for insured and uninsured deposit flows in Q4-2022, supply-driven deposit inflows in 2020, and measures of 2022 stock returns and the value of the deposit franchise, as in Table 4. Details on variable construction and data sources are provided in Appendix A.1. Robust standard errors are reported in parentheses.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

- **Total outflows.** The same effect is weaker for total deposits, which is intuitive because uninsured deposits are the most run-prone.
- **Punchline.** Social media does not just predict equity-market reactions; it is also linked to actual deposit flight.

5.5 Table 6 and Figure 8: hourly attention during the run

- **Read.** Table 6 estimates hourly return regressions with lagged Twitter attention.
- **Low run risk banks.** The interaction between post-March-9 and high lagged attention is small and insignificant.
- **High run risk banks.** The same interaction is negative and significant, around **-0.096** to **-0.100** basis points per hour in the full-window fixed-effects specifications.
- **Short window.** Around the immediate onset of the run, the magnitude becomes much larger, around **-0.147** to **-0.214**.
- **Figure 8.** Cumulative returns following high versus low Twitter attention diverge only in the high-run-risk subsample, and only once the run starts.

5.6 Table 7: retweets and alternative attention windows

- **Different lags.** The results are similar whether attention is measured over the past **1 hour**, **2 hours**, or **4 hours**. So the findings are not an artifact of a specific lag definition.

Table 6
Social media and the bank run — hourly frequency.

Panel A: Run risk and Tweet volume over the prior 4 h

	Hourly stock return					
	Low run risk			High Run Risk		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.201***	−0.209***		−0.150***	−0.165***	
	(0.019)	(0.020)		(0.051)	(0.046)	
1(# Tweets (4 h) High) (t − 1)	−0.019**	0.007	0.007	−0.009	0.033**	0.031***
	(0.009)	(0.012)	(0.010)	(0.006)	(0.013)	(0.011)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	0.007	0.017	0.016	−0.096**	−0.108**	−0.100**
	(0.029)	(0.029)	(0.027)	(0.038)	(0.047)	(0.042)
Ret (Cum.-4 h) (z) (t − 1)	0.052	0.032	0.023	0.096*	0.042	0.028
	(0.036)	(0.036)	(0.046)	(0.056)	(0.044)	(0.068)
1(≥ Mar 09) × Ret (Cum.-4 h) (z) (t − 1)	−0.107***	−0.095***	−0.107***	−0.003	0.011	0.031
	(0.038)	(0.034)	(0.031)	(0.073)	(0.077)	(0.075)
N Articles (4 h) (z) (t − 1)	−0.001*	−0.003***	−0.001**	0.029**	0.221	0.146
	(0.000)	(0.000)	(0.000)	(0.013)	(0.148)	(0.147)
1(≥ Mar 09) × N Articles (4 h) (z) (t − 1)	0.006***	0.005***	0.007***	−0.176**	−0.014	−0.031
	(0.001)	(0.001)	(0.002)	(0.083)	(0.121)	(0.099)
Constant	−0.030***			−0.038***		
	(0.004)			(0.004)		
Observations	15,330	15,330	15,330	14,996	14,996	14,996
R^2	0.013	0.019	0.236	0.022	0.040	0.245
Within R^2		0.013	0.004		0.016	0.003
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

Panel B: Short- vs. long event window

	Hourly stock return					
	March 08–09			March 06–10		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.589***	−0.677***		−0.150***	−0.165***	
	(0.169)	(0.162)		(0.051)	(0.046)	
1(# Tweets (4 h) High) (t − 1)	−0.008	0.036***	0.025**	−0.009	0.033**	0.031***
	(0.009)	(0.013)	(0.011)	(0.006)	(0.013)	(0.011)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	−0.186**	−0.214**	−0.147***	−0.096**	−0.108**	−0.100**
	(0.075)	(0.085)	(0.054)	(0.038)	(0.047)	(0.042)
Constant	−0.021***			−0.038***		
	(0.007)			(0.004)		
Observations	5577	5577	5577	14,996	14,996	14,996
R^2	0.162	0.213	0.359	0.022	0.040	0.245
Within R^2		0.155	0.053		0.016	0.003
Return Controls + Interactions	✓	✓	✓	✓	✓	✓
News Controls + Interactions	✓	✓	✓	✓	✓	✓
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

Panel C: Excluding SIVB from the sample

	Hourly stock return					
	Short Window, Excl. SIVB			Long Window, Excl. SIVB		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.777***	−0.842***		−0.199***	−0.216***	
	(0.078)	(0.080)		(0.021)	(0.022)	
1(# Tweets (4 h) High) (t − 1)	−0.003	0.028*	0.020	−0.007	0.028**	0.028**
	(0.008)	(0.016)	(0.013)	(0.006)	(0.013)	(0.011)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	−0.184*	−0.203*	−0.152**	−0.091**	−0.081*	−0.079*
	(0.098)	(0.107)	(0.076)	(0.043)	(0.045)	(0.043)
Constant	−0.001			−0.036***		
	(0.007)			(0.004)		
Observations	5538	5538	5538	14,891	14,891	14,891
R^2	0.192	0.226	0.379	0.016	0.024	0.269
Within R^2		0.210	0.008		0.018	0.013
Return Controls + Interactions	✓	✓	✓	✓	✓	✓
News Controls + Interactions	✓	✓	✓	✓	✓	✓
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

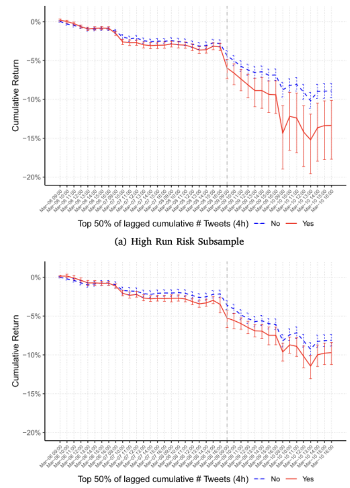
(continued on next page)

Table 6 (*continued*).

Panel D: Retweet (RT)-weighted Tweets

	Hourly stock return		
	High run risk		
	(1)	(2)	(3)
1(≥ Mar 09)	−0.160***	−0.176***	
	(0.055)	(0.050)	
1(# RT-weighted Tweets (4 h) High) (t − 1)	−0.008	0.035**	0.034***
	(0.006)	(0.015)	(0.012)
1(≥ Mar 09) × 1(# RT-weighted Tweets (4 h) High) (t − 1)	−0.078**	−0.090**	−0.082**
	(0.033)	(0.043)	(0.038)
Ret (Cum.-4 h) (z) (t − 1)	0.098*	0.043	0.030
	(0.056)	(0.044)	(0.068)
1(≥ Mar 09) × Ret (Cum.-4 h) (z) (t − 1)	−0.004	0.010	0.029
	(0.074)	(0.077)	(0.076)
N Articles (4 h) (z) (t − 1)	0.030**	0.223	0.147
	(0.013)	(0.148)	(0.147)
1(≥ Mar 09) × N Articles (4 h) (z) (t − 1)	−0.179**	−0.018	−0.036
	(0.084)	(0.122)	(0.100)
Constant	−0.039***		
	(0.004)		
Observations	14,786	14,786	14,786
R^2	0.022	0.040	0.246
Within R^2		0.016	0.003
Firm FE		✓	✓
Day-by-Hour FE			✓

This table presents OLS regression estimates that relate hourly bank stock returns to lagged Twitter attention, interacted with an indicator for the onset of the SVB run on March 9. The dependent variable in all columns is the hourly return (in basis points) for a bank stock in our sample. The sample is organized at the firm-by-day-by-hour level. We split the sample into below-median (columns 1–3) and above-media (columns 4–6) banks with respect to "Run Risk (z)", which is defined as '% Uninsured' × % Loss MTM' and estimate the regression separately for the two subsamples. The indicator variable '1(≥ Mar 9)' equals one after the onset of the SVB bank run on March 9, 2023, at 9am, and zero otherwise. Firm- and Day-by-Hour fixed effects are included as indicated. Lagged Twitter attention is an indicator variable denoted as '1(# Tweets (4 h) High) (t − 1),' which takes the value of 1 if the number of Tweets posted about the bank in the 4 hours prior to the current period t (i.e., hours $t-4$ through $t-1$) was above the median, and zero otherwise. We additionally control for the number of news articles posted about a bank, the bank's cumulative stock return in hours $t-5$ through $t-1$, and the interactions between these variables with the run period indicator in all columns. Panel A presents the baseline estimates. Panel B employs a shorter sample around from March 08 (9am) to March 09 (4pm) in columns 1–3. Panel C excludes Silicon Valley Bank (SIVB) from the sample, and Panel D uses the number of tweets weighted by the corresponding number of Retweets (within the next hour) to measure social media activity. Standard errors reported in parentheses are clustered at the bank level.
* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.



(a) High Run Risk Subsample

(b) Low Run Risk Subsample

Fig. 8. *Cumulative Returns Following High versus Low Twitter Attention.* This figure shows the average cumulative returns of publicly listed banks and bank holding companies (BHCs) from March 06 until March 10, 2023 (during trading hours), splitting the sample into banks with high run risk (Panel (a)) and low run risk (Panel (b)). In each subsample we split the sample of bank stocks into those with above (red solid line) and below (blue dashed line) median number of tweets over the previous four hours, calculate average returns for the two groups, and cumulate the average returns at the hourly frequency from March 06 until March 10, 2023. The gray vertical dashed line indicates March 09, 2023 at 9am. 95% confidence intervals around the mean cumulative return are constructed by bootstrapping with clustering at the bank level.

- **Retweet heterogeneity.** The attention effect is much stronger when recent tweets receive many retweets. In the fixed-effects specifications, the key interaction is about **-0.124** in the high-retweet subsample versus about **-0.075** in the low-retweet subsample.
- **Punchline.** What matters is not just raw tweet volume, but socially propagated tweet volume.

Table 7
Hourly frequency — different lags and tweet attention.

Panel A: Baseline specification with different lags

	Hourly stock return		
	1 h	2 h	4 h
	(1)	(2)	(3)
1(# Tweets High) (t − 1)	0.038**	0.028*	0.031***
	(0.018)	(0.015)	(0.011)
1(≥ Mar 09) × 1(# Tweets High) (t − 1)	−0.074*	−0.084*	−0.100**
	(0.045)	(0.046)	(0.042)
Observations	14,996	14,996	14,996
R^2	0.244	0.244	0.245
Within R^2	0.003	0.003	0.003
Return Controls + Interactions	✓	✓	✓
News Controls + Interactions	✓	✓	✓
Firm FE	✓	✓	✓
Day-by-Hour FE	✓	✓	✓

Panel B: Tweets with high- vs low number of Retweets

	Hourly stock return					
	Retweets low			Retweets High		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.226***	−0.228***		−0.133**	−0.164***	
	(0.043)	(0.046)		(0.066)	(0.062)	
1(# Tweets (4 h) High) (t − 1)	0.005	0.047***	0.021*	−0.019**	0.006	0.035**
	(0.011)	(0.015)	(0.013)	(0.009)	(0.019)	(0.014)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	−0.037	−0.047	−0.075*	−0.141***	−0.179***	−0.124**
	(0.051)	(0.065)	(0.039)	(0.053)	(0.068)	(0.062)
Constant	−0.036***			−0.033***		
	(0.006)			(0.006)		
Observations	7485	7485	7485	7511	7511	7,511
R^2	0.037	0.052	0.391	0.021	0.056	0.225
Within R^2		0.035	0.024		0.016	0.003
Return Controls + Interactions	✓	✓	✓	✓	✓	✓
News Controls + Interactions	✓	✓	✓	✓	✓	✓
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

This table presents OLS regressions that relate hourly stock return (in bps) to lagged in-run Twitter attention measures, analogous to Table 6. We include only the subset of banks in the high-run risk subsample. Panel A presents specifications similar to these but considering tweet counts over the previous 1 h, 2 h, and 4 h, respectively, to construct '1(# Tweets High) (t − 1)' across columns 1–3. Panel B further splits the sample into observations with a high (above-median) and low (below-median) number of Retweets (RTs), and estimates the model separately for the two subsamples. Stock return and newspaper controls and their interactions with '1(≥ Mar 09)' as well as bank and day-by-hour fixed effects are included as indicated. Standard errors reported in parentheses are clustered at the bank level.
* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.

Table 8
High frequency return responses to tweet sentiment.

	(1) $\Delta p_{i,t}$	(2) $\Delta p_{i,t}$	(3) $\Delta p_{i,t}$	(4) $\Delta p_{i,t}$	(5) $\Delta p_{i,t}$	(6) $\Delta p_{i,t}$	(7) $\Delta p_{i,t}$	(8) $\Delta p_{i,t}$	(9) $\Delta p_{i,t}$	(10) $\Delta p_{i,t}$
VADER Pos(z)	−1.83*	−1.70	−1.39	−1.31	0.76	1.05	−1.82*	−1.75	0.77	1.14
	(1.07)	(1.22)	(1.25)	(1.40)	(1.05)	(1.17)	(1.04)	(1.20)	(0.94)	(1.07)
VADER Neg(z)	2.64	3.04	2.93	3.34*	−0.18	0.18	3.07	3.49*	1.65	1.97
	(2.01)	(1.85)	(2.04)	(1.95)	(1.42)	(1.41)	(2.19)	(2.02)	(1.56)	(1.55)
Tech Community Flag	13.35*	12.64							13.34*	12.70
	(7.86)	(7.81)							(7.88)	(7.81)
VADER Pos(z) × Tech Community Flag	3.59	1.67							4.10	2.25
	(7.36)	(7.55)							(7.12)	(7.24)
VADER Neg(z) × Tech Community Flag	−12.03***	−11.14**							−11.34**	−10.44*
	(4.56)	(5.45)							(4.79)	(5.70)
Run Tweet			−1.31	−1.38					−2.22	−2.21
			(4.40)	(4.56)					(4.60)	(4.69)
VADER Pos(z) × Run Tweet			−5.42	−6.08					−6.14	−6.64
			(9.04)	(10.17)					(9.10)	(10.16)
VADER Neg(z) × Run Tweet			−13.27***	−12.91***					−12.73***	−12.35***
			(2.93)	(2.81)					(2.69)	(2.76)
VADER Pos(z) × High Run Risk (M2)					−2.97*	−3.26*			−3.05*	−3.31*
					(1.59)	(1.81)			(1.62)	(1.83)
VADER Neg(z) × High Run Risk (M2)					2.99	3.07			3.08	3.15
					(2.72)	(2.57)			(2.87)	(2.72)
Contagion Tweet							15.37	11.27	16.44	12.34
							(26.32)	(25.70)	(26.54)	(26.01)
VADER Pos(z) × Contagion Tweet							18.99	17.33	18.55	17.02
							(12.36)	(12.08)	(12.54)	(12.20)
VADER Neg(z) × Contagion Tweet							−22.36**	−20.63*	−22.63**	−20.91*
							(11.00)	(11.19)	(11.14)	(11.35)
$p_{i,t-\epsilon} - p_{i,t-10}$		0.08**		0.08**		0.08**		0.08**		0.08**
		(0.03)		(0.03)		(0.03)		(0.03)		(0.03)
Total News (−10, −ϵ) (z)		0.77		0.70		0.75		0.82		0.78
		(3.58)		(3.67)		(3.59)		(3.61)		(3.63)
ESS (−10, −ϵ) (z)		17.92***		17.90***		17.93***		17.98***		17.93***
		(4.15)		(4.19)		(4.14)		(4.12)		(4.16)
Constant	−15.29***	−11.59***	−14.57***	−10.89***	−14.82***	−11.15***	−14.53***	−10.84***	−14.80***	−11.08***
	(4.04)	(2.42)	(4.03)	(2.37)	(3.84)	(2.28)	(3.92)	(2.42)	(4.18)	(2.55)
Observations	36751	36659	36751	36659	36751	36659	36751	36659	36751	36659
R^2 (%)	1.07	2.5	1.06	2.5	1.05	2.49	1.08	2.51	1.11	2.54
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09

This table presents OLS estimates for how high frequency returns from *just* before the tweet to 5 minutes after relate to tweet sentiment interacted with whether the tweet is run-relevant ("run" or "contagion" tweets, or authored by tech community users) and whether the bank has high run risk. The level of observation is at the individual tweet level. In all regressions, the dependent variable $\Delta p_{i,t}$ is the log change in bank i's price between the last trade before the Tweet was posted and the first trade in the [+5, +15] minute window around the tweet at time t. $\Delta p_{i,t}$ is winsorized at the 1% level and expressed in basis points. VADER Pos (z) and VADER Neg (z) are the positive and negative components of the VADER sentiment respectively standardized using the sample used in each regression. 'Tech Community Flag' indicates that a given tweet was posted by a member of the tech community, as constructed in Section Section 2.1.2. 'Contagion Tweet' and 'Run Tweet' indicate if the tweet contains at least one token in the contagion and run behavior dictionaries respectively. 'High Run Risk' indicates if the bank is in the top quintile of bank run risk (i.e., MTM losses × uninsured deposits). We exclude observations for which the traded volume associated with the last price in the window before the tweet or the first price in the window after the tweet is zero. In all regressions, we include bank fixed effects. Specifications in even columns additionally control for the bank stock return in the 10 minutes prior to the tweet ('$p_{i,t-\epsilon} - p_{i,t-10}$') We also control for the number of articles posted in the 10 minutes before the Tweet ('Total News') and the corresponding news sentiment ('ESS') using data from RavenPack. Standard errors are reported in parenthesis and clustered at the bank-day level.
* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.

Table 9
High frequency tests, Controlling for retweets and followers.

	(1) $\Delta p_{i,t}$	(2) $\Delta p_{i,t}$	(3) $\Delta p_{i,t}$	(4) $\Delta p_{i,t}$	(5) $\Delta p_{i,t}$	(6) $\Delta p_{i,t}$	(7) $\Delta p_{i,t}$	(8) $\Delta p_{i,t}$
VADER Pos(z)	−1.70	−2.49	−1.31	−2.13	−1.75	−2.56	1.05	0.03
	(1.22)	(1.76)	(1.40)	(1.99)	(1.20)	(1.79)	(1.17)	(1.53)
VADER Neg(z)	3.04	2.71	3.34*	3.13*	3.49*	3.07*	0.18	0.06
	(1.85)	(1.79)	(1.95)	(1.88)	(2.02)	(1.84)	(1.41)	(1.89)
Tech Community Flag	12.64	13.55*						
	(7.81)	(7.68)						
VADER Pos(z) × Tech Community Flag	1.67	0.68						
	(7.55)	(7.87)						
VADER Neg(z) × Tech Community Flag	−11.14**	−11.38**						
	(5.45)	(5.53)						
Run Tweet			−1.38	−0.20				
			(4.56)	(4.08)				
VADER Pos(z) × Run Tweet			−6.08	−8.10				
			(10.17)	(10.86)				
VADER Neg(z) × Run Tweet			−12.91***	−13.81***				
			(2.81)	(3.06)				
Contagion Tweet					11.27	11.41		
					(25.70)	(25.90)		
VADER Pos(z) × Contagion Tweet					17.33	16.59		
					(12.08)	(11.67)		
VADER Neg(z) × Contagion Tweet					−20.63*	−20.62*		
					(11.19)	(11.10)		
VADER Pos(z) × High Run Risk (M2)							−3.26*	−3.02*
							(1.81)	(1.78)
VADER Neg(z) × High Run Risk (M2)							3.07	2.84
							(2.57)	(2.44)
High Followers (M2)		−1.25		−0.94		−0.99		−0.89
		(3.72)		(3.86)		(3.74)		(3.70)
VADER Pos(z) × High Followers (M2)		0.44		0.52		0.43		0.44
		(3.17)		(3.10)		(3.11)		(3.17)
VADER Neg(z) × High Followers (M2)		1.22		0.95		1.38		1.13
		(2.84)		(2.76)		(2.98)		(2.80)
High Retweets (M2)		−10.67		−10.43		−9.99		−10.02
		(8.56)		(8.07)		(8.50)		(8.49)
VADER Pos(z) × High Retweets (M2)		12.59***		14.15***		12.65***		12.70***
		(3.43)		(4.41)		(3.45)		(3.37)
VADER Neg(z) × High Retweets (M2)		−1.52		−1.26		−1.34		−1.68
		(6.43)		(6.67)		(6.73)		(6.44)
$p_{i,t-\epsilon} - p_{i,t-10}$	0.08**	0.08**	0.08**	0.08**	0.08**	0.08**	0.08**	0.08**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Total News (−10, −ϵ) (z)	0.77	0.78	0.70	0.69	0.82	0.83	0.75	0.76
	(3.58)	(3.62)	(3.67)	(3.70)	(3.61)	(3.64)	(3.59)	(3.62)
ESS (−10, −ϵ) (z)	17.92***	17.95***	17.90***	17.90***	17.98***	18.00***	17.93***	17.95***
	(4.15)	(4.18)	(4.19)	(4.23)	(4.12)	(4.15)	(4.14)	(4.18)
Constant	−11.59***	−10.31***	−10.89***	−9.77***	−10.84***	−9.68***	−11.15***	−10.01***
	(2.42)	(3.25)	(2.37)	(3.37)	(2.42)	(3.27)	(2.28)	(3.16)
Observations	36 659	36 659	36 659	36 659	36 659	36 659	36 659	36 659
R^2 (%)	2.5	2.52	2.5	2.52	2.51	2.53	2.49	2.51
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09

This table presents OLS estimates for how high frequency returns relate to tweet sentiment interacted with whether the tweet is run-relevant ("run" or "contagion" tweets, or authored by tech community users) and whether the bank has high run risk, *controlling for measures of tweet influence*. The specifications and variable definitions are identical to Table 8, except that — in even columns — we additionally control for 'High Followers' and 'High Retweets,' which indicate if the author of the tweet is in the top quintile number of followers and if the Tweet received top quintile number of Retweets, respectively. Standard errors are reported in parenthesis and doubled clustered at the bank-day level.
* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.

5.7 Table 8 and Table 9: tweet-level market impacts

- **Read.** These tables ask which kinds of tweets move prices in the minutes right after they are posted.
- **Tech-community tweets.** Negative sentiment in tech-community tweets has a strong immediate effect, around **-11 to -12 basis points**.

- **Run tweets.** Negative sentiment in run tweets has an even stronger effect, around **-13 basis points**.
- **Contagion tweets.** Negative sentiment in contagion tweets has the largest effect, around **-20 to -23 basis points**.
- **High run risk interaction.** The interaction between negative sentiment and high run risk is small and insignificant.
- **Table 9.** Adding controls for follower counts and retweets does not materially change the main interaction estimates.
- **Punchline.** Run-relevant communication moves prices immediately, but negative sentiment per se does *not* explain why some banks are more run-prone than others. That role belongs to attention and coordination.

6 Short summary

- **Method.** Measure ex ante Twitter exposure before the SVB collapse, then relate it to run-period stock losses, deposit outflows, and high-frequency return reactions.
- **Identification.** Use pre-run attention, weak correlation between pre-exposure and run risk, rich controls, lagged hourly attention, and minute-level event windows around tweets.
- **Application.** U.S. publicly traded banks and bank holding companies during the March 2023 banking stress episode.
- **Main result.** Social media **amplifies classical bank run risks**. Attention and run-relevant communication matter; negative sentiment alone does not.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that banks with greater pre-existing exposure to Twitter experienced a more severe run in March 2023.
- This was especially true for banks with the classic fragilities of **high uninsured deposits** and **large mark-to-market losses**.
- The evidence is strongest for **Twitter attention** and the social propagation of run-relevant messages, not for sentiment-driven pessimism alone.

7.2 Economic intuition

- In a fragile banking system, uninsured depositors must coordinate to run.
- Social media lowers the cost and increases the speed of that coordination.

- A bank that is already salient on social media becomes easier to focalize during stress, so once bad news hits, attention can quickly turn into pressure.
- The key mechanism is therefore not that Twitter reveals hidden insolvency with great precision, but that it helps convert existing fragility into realized distress.

7.3 Takeaway

This paper's main lesson is that social media should be treated as part of the modern bank-run technology. Traditional balance-sheet weaknesses still determine which banks are fragile, but social media can accelerate and magnify the pressure on those banks by concentrating depositor attention and communication exactly when coordination matters most.